

USB CHARGERS

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45-Watt USB Charger



This reference design is for a USB power delivery (PD) 45W mobile charger that can cater to rechargeable devices with power requirements of up to 100W. The USB PD charger functions with an AC mains voltage ranging from 90V (AC) to 264V (AC).

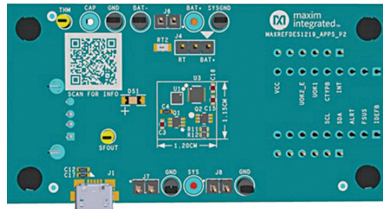
The TEA1936xDB1533 is a reference design by NXP, that provides a maximum output power of 45 W, with a maximum current of 3A. The design incorporates the quasi-resonant controller, synchronous rectifier controller, and controllers for USB-PD Type-C and Quick Charge (QC) 4.0. The design caters to power supplies for mobile and notebook devices, supporting up to 100W power requirements. It offers the flexibility to select output voltages ranging from 5V to 20V. The design supports multiple protocols, including USB Power Delivery (USB-PD) 2.0 and 3.0, as well as Quick Charge 4.0. It incorporates features such as low no-load power, minimal audible noise, and low output voltage ripple, ensuring excellent performance for the USB PD charger. The USB PD charger functions with an AC mains voltage of 90V (AC) to 264V (AC) and delivers a maximum power output of 45 watts. With an ambient temperature range of -40°C to 85°C inside the casing, the design operates efficiently within an input frequency range of 47Hz to 63Hz, achieving remarkable efficiency exceeding 92%.

The reference design meets the power supply needs of mobile and notebook devices, supporting power requirements of up to 100W.

NXP

<https://www.electronicsforu.com/electronics-projects/industry-grade-projects/reference-design-for-a-45-watt-usb-charger>

USB BC 1.2 Compliant Charger



Reference design for a USB battery charging (BC) 1.2 compliant charger that provides up to 2A of charging current. The charger is capable of detecting different battery charging methods, including Standard Downstream Port (SDP), Charging Downstream Port (CDP), and Dedicated Charging Port (DCP).

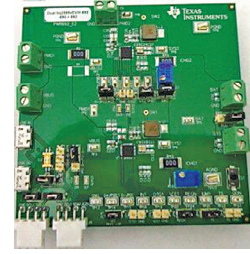
The reference design of a single cell Li-ion battery charger by Maxim Integrated provides up to 2A of charging current. The reference design adheres to USB Battery charging Rev1.2* standards. The charger can detect various battery charging methods such as Standard Downstream Port (SDP), Charging Downstream Port (CDP), and Dedicated Charging Port (DCP). Additionally, the reference design can identify common proprietary charge adaptors, including those from Apple1. The reference design utilises the ModelGauge m5 algorithm, known for its accurate estimation and low power consumption. It is suitable for a wide range of applications, including portable consumer devices, digital imaging (DSC, DVC), portable industrial devices, and USB 1-cell pack equipment, among others. The design provides a user-friendly graphical user interface (GUI) for effortless monitoring and control of the parameters. The software is compatible with Windows 10, facilitating easy integration and operation. The design also includes an I2C serial interface for convenient communication and connectivity.

The reference design can be used for various applications, including portable consumer devices, digital imaging devices, portable industrial devices, USB 1-cell pack equipment, etc.

Maxim Integrated

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Smartphone Fast Charger Reference Design



The reference design is for a single-cell Li-ion charger that supports a charging current of up to 5A when powered by an optimal 12V source. The design demonstrates a maximum efficiency of up to 94%. The device is compliant with USB 2.0 and USB 3.0 power specifications.

TIDA00590 is the reference design for a dual battery charger by Texas Instruments (TI), which uses a dual charging IC configuration to achieve a fast charge with low heat dissipation. When powered by an optimal 12V source, the charger can deliver a charging current of up to 5A while maintaining a low thermal budget. The reference design has an efficiency of up to 94%. The dual chargers surpass their 5A maximum charge currents, ensuring an efficient distribution of heat loss across the board. The module incorporates a low impedance power path to enhance efficiency, resulting in reduced battery charging time and extended battery life during the discharging phase. The design exhibits an efficiency exceeding 90%, with a maximum efficiency of over 94%. The device is compatible with various input sources, including a standard USB host port, USB charging port, and USB-compliant adjustable high voltage adaptor. The design is compliant with USB 2.0 and USB 3.0 power specifications, incorporating input current and voltage regulation.

The reference design is suitable for portable electronic devices, mobile accessories, industrial equipment, medical devices, automotive systems, energy storage, and IoT devices, etc.

Texas Instruments (TI)

<https://www.electronicsforu.com/electronics-projects/industry-grade-projects/smartphone-fast-charger-reference-design>